PH.D. DISSERTATION DEFENSE

An Automated 3D Mapping and Point Cloud Semantic Modeling Framework for Spatial-Operation Robots

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[Laboratory name] | [Defense date]

Outline

01 Introduction

02 Related Work

03 Proposed Framework

04 Experiments and Results

05 Conclusion and Future Work

01

Introduction

Motivation, problem, and research objectives

Research Background

- [Application context]
- [Current limitation]
- [Why this matters]

CONTEXT FIGURE

[Replace with one clear image]

[One sentence explaining the research need]

Problem Definition and Objectives

Problem

- [Limitation 1]
- [Limitation 2]
- [Limitation 3]

Objectives

- [Objective 1]
- [Objective 2]
- [Objective 3]

[Define the scope of this dissertation]

02

Related Work

Position the research and identify the gap

Comparison of Related Work

Approach	Strength	Remaining gap
[Method A]	[Strength A]	[Gap A]
[Method B]	[Strength B]	[Gap B]
[Method C]	[Strength C]	[Gap C]

[The unresolved gap addressed by this work]

Research Gap

01 [Gap in acquisition]

02 [Gap in semantic modeling]

03 [Gap in downstream use]

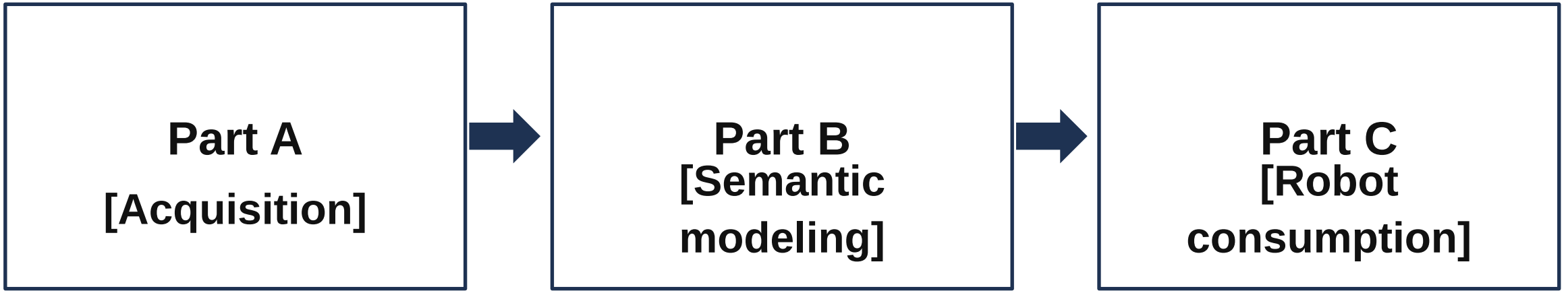
03

Proposed Framework

Present the core idea and method

Framework Overview

[One integrated idea connecting the three parts]



[State the principal contribution]

Part A — Automated Mapping

METHOD DIAGRAM

[Insert acquisition workflow]

Key mechanism

- **[Input]**
- **[Decision rule]**
- **[Output]**

[What is new in this component?]

Part B — Semantic Modeling

METHOD DIAGRAM

[Insert semantic workflow]

Key mechanism

- **[Geometry]**
- **[Verification]**
- **[Graph update]**

[What is new in this component?]

Part C — Digital-Twin Integration

SYSTEM DIAGRAM

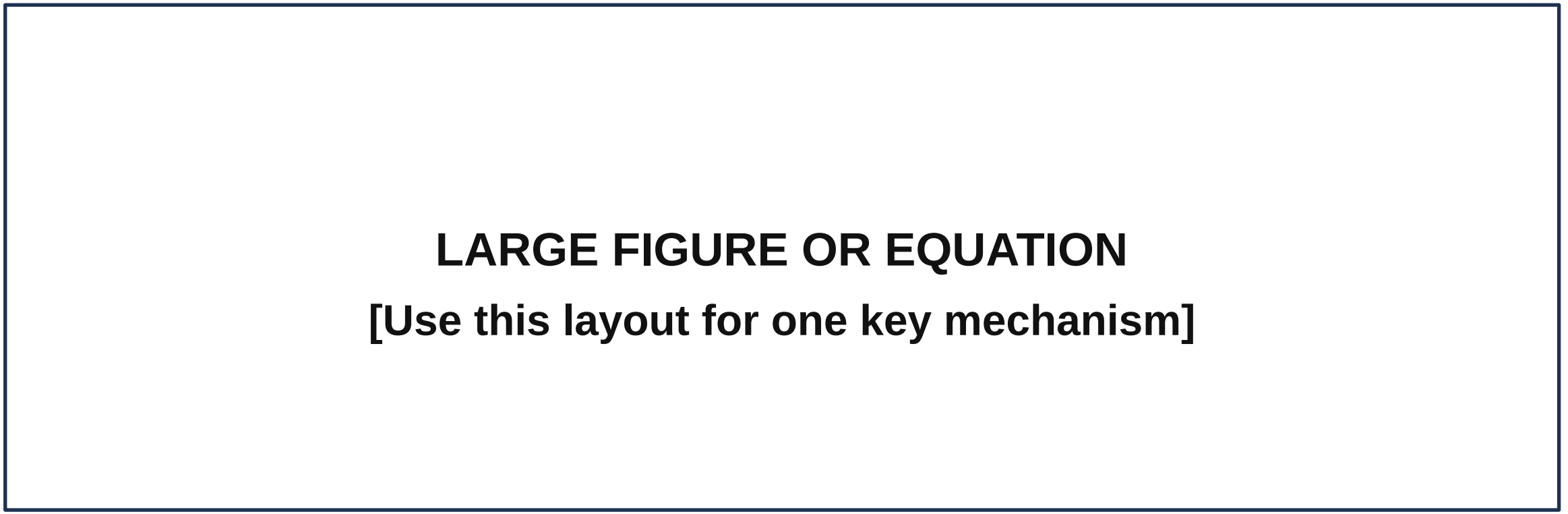
[Insert twin and robot interfaces]

Key mechanism

- **[Shared model]**
- **[Frame alignment]**
- **[Mission interface]**

[State the demonstrated integration scope]

[Method Detail Title]



[One sentence interpreting the figure]

04

Experiments and Results

Show evidence, controls, and limitations

Experimental Setup

- [Platform and sensor]
- [Environment or dataset]
- [Metric and comparison]

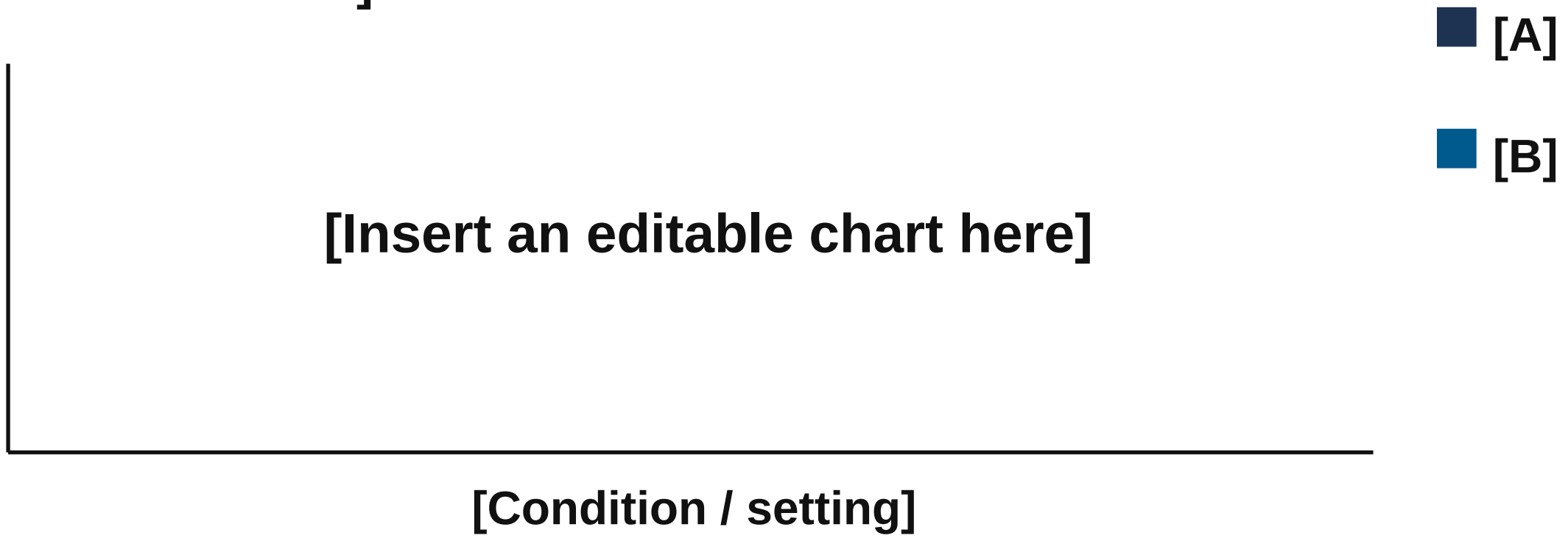
SETUP FIGURE

[Replace with experiment image]

[State the test scope and sample size]

Quantitative Results

[Metric and unit]



[Report the main finding with its uncertainty]

Qualitative Results

[Baseline]



[Proposed]



[Describe one visible difference under matched conditions]

Ablation Study

Configuration	Metric 1	Metric 2
[Baseline]	—	—
[+ Component A]	—	—
[+ Component B]	—	—

[Identify the contribution isolated by this experiment]

Discussion and Limitations

Supported findings

- [Finding 1]
- [Finding 2]
- [Finding 3]

Remaining limits

- [Limit 1]
- [Limit 2]
- [Limit 3]

[State how far the evidence supports the claim]

05

Conclusion and Future Work

Close with contributions and next steps

Summary of Contributions

01 [Contribution to automated mapping]

02 [Contribution to semantic modeling]

03 [Contribution to robot integration]

Future Work

- [Method extension]
- [Independent validation]
- [Deployment study]

Questions and Discussion

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